

# Learning Personalized Photographic Style from Pairwise User Preferences

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## Abstract

*Photographic style preferences are deeply personal, varying across individuals in color and tonal aesthetics. We introduce Personalized Photographic Style (PPS) learning, where the goal is to capture a user’s implicit preferences from comparative judgments and apply them consistently across diverse images. To establish a foundation for this problem, we present three contributions. First, we introduce PPSD, a dataset containing pairwise preference judgments from 767 users, each providing an average of 70 comparisons. To capture diverse style signals, images are sourced from professional edits, device pipelines, and generative models. Second, we explore several baseline models demonstrating the feasibility of adapting style transfer and enhancement approaches for preference learning. Third, we develop a comparative evaluation framework suited to the implicit nature of personal preferences. Dataset is available at <https://hello-jinwoo.github.io/pps>.*

## 1. Introduction

The pursuit of a “good photo” is a universal yet deeply personal endeavor [4, 17, 30, 46]. At its core, this pursuit is about aligning an image with a user’s subjective aesthetic taste without a single ground truth, particularly in the dimensions of color and tone. While manual editing programs like Adobe Lightroom offer various control tools, they require significant expertise. This leaves a gap for automated systems that can learn and apply a user’s unique visual preferences. We term this challenge Personalized Photographic Style (PPS) learning, the task of capturing and reproducing an individual’s aesthetic preferences across diverse photographic contexts.

The field has explored related problems through various approaches. Photo-realistic Style Transfer (PST) [11, 12, 18, 22, 27, 34, 38], for example, aims to transfer the color and tonal characteristics from a reference image to a target image. PST enables personalization by allowing

users to select a preferred reference image as a style template. The target style is thus explicitly determined by the user’s choice of reference. In another direction, Personalized Image Enhancement (PIE) [9, 26, 28, 32, 40] focuses on learning user-specific enhancement preferences to improve image quality. These approaches work with well-defined source-target pairs, such as learning mappings from specific photographer styles [8, 35] or enhancing synthetically degraded images [32]. In both paradigms, the source and target color style are relatively well-defined, enabling structured transformations that, while effective in controlled settings, do not readily extend to diverse real-world retouching scenarios.

Given this limitation, PPS learning presents a distinct challenge. The goal is to learn a user’s aesthetic preference from their comparative preferences and apply it to retouch new photographs in their preferred color and tonal style. Specifically, the model observes a user’s choices between different tonal renderings of the same scene to infer their implicit aesthetic, then generalizes this to new photographs with different content. Unlike PST or PIE, there is no single reference image to mimic and no explicit degradation to reverse. Instead, the learning signal comes solely from observing which photographic styles a user prefers when the same scene is rendered with different color and tonal adjustments. The target aesthetic exists only implicitly in the user’s mind, revealed gradually through these relative choices across diverse images.

Despite growing interest in personalization [48, 50, 51, 53], progress in PPS has been hindered by three key challenges: the absence of large-scale data, a limited understanding of effective methods for ambiguous preference learning, and the lack of a proper evaluation framework.

To this end, we make three core contributions. First, we introduce the PPSD (Personalized Photographic Style Dataset), a large-scale collection of real-world pairwise user preferences. The dataset contains responses from 767 users (avg. 70 comparisons each), with image pairs sourced from diverse professional edits, device pipelines, and generative models. To our knowledge, this is the first dataset of its scale and diversity for PPS learning. Second, we conduct

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\*Work done during an internship at CIPLAB.

an empirical study with several representative approaches, demonstrating the feasibility of adapting existing models for PPS learning and providing a foundation for future research. Third, we develop a comparative evaluation framework suited to the implicit, comparative nature of personal preferences, for which conventional metrics are inadequate.

By introducing PPSD along with baseline methods and evaluation frameworks, we provide the foundation to explore this emerging problem. Additionally, we discuss key considerations for future work in learning and applying personalized photographic styles. We hope this work contributes to the development of imaging systems that better adapt to individual aesthetic preferences.

## 2. Related Work

### 2.1. Photo-realistic Style Transfer

Photo-realistic style transfer (PST) aims to transfer the color and tonal characteristics from a reference image to a target image while preserving the target’s structure and content. Common approaches include feature transformation techniques [13, 18, 34], predicting parameters for traditional image filters [21, 27, 43], and 3D/4D LUT-based methods for real-time transfer [19, 33, 59].

A defining characteristic of PST is its reference-based nature, where style is explicitly specified through a reference image. While this allows preference expression through reference selection, personal aesthetic preferences are difficult to capture through single examples. PPS addresses this by learning from multiple preference comparisons across diverse contexts, enabling automated personalization without per-image reference specification.

### 2.2. Personalized Image Enhancement

Personalized image enhancement (PIE) focuses on learning user-specific or photographer-specific image enhancement. Early work explored learning editing styles from user interactions [8] and inferring adjustments from before-after pairs [23]. Recent methods such as PieNet [28] learn user-specific preference vectors from limited data, while PIE-MSM [32] extends this with masked style modeling using a Transformer to predict content-aware styles.

A common characteristic of PIE methods is that they operate with well-defined learning signals, whether through photographer identities (e.g., Expert A or B in Adobe5K dataset) or reference processing pipelines in quality enhancement tasks. PPS operates differently in that the desired aesthetic exists only implicitly in the user’s mind and must be inferred from comparative judgments.

### 2.3. Comparative Preference Datasets

Comparative preference datasets have emerged as a valuable resource for learning from human feedback. In natural

language processing, datasets like Anthropic HH-RLHF [5] and UltraFeedback [15] provide pairwise preferences for language model responses. In computer vision, Pick-a-Pic [31] and HPDv3 [41, 55, 56] collect large-scale pairwise preferences for text-to-image generation from real users.

The comparative approach offers key advantages for capturing subjective preferences, as pairwise comparisons are more natural and reliable than absolute ratings when no objective ground truth exists. PPSD adopts this framework to capture personalized preferences in photographic style. While existing vision preference datasets focus on content-related aspects in text-to-image generation, PPSD addresses the perceptual domain of color and tone, where individual aesthetic judgments are particularly subtle and subjective.

### 2.4. Datasets for Photo-realistic Image Retouching

Several public datasets have advanced photo-realistic image retouching. Adobe-MIT 5K [8] provides 5,000 raw images with edits from five professional photographers, enabling photographer-specific style learning. PPR10K [35] offers over 10,000 portrait photos with expert retouching and group-level style consistency. PIE-MSM [32] uses Flickr images to create pseudo source-target pairs through learned degradation, supporting unpaired learning scenarios. These datasets provide source-to-target scenarios for retouching research. PPSD builds on these foundations, along with open-source image services (e.g., Pixabay, Unsplash) and generative models, to collect personalized preferences through comparative judgments.

## 3. Personalized Photographic Style Dataset

### 3.1. Dataset Overview

We introduce the Personalized Photographic Style Dataset (PPSD), a large-scale collection of pairwise user preferences for photographic style. Following previous image preference datasets [31, 41], we build our dataset on pairwise comparisons rather than absolute ratings, as photographic style is inherently perceptual and users can more readily identify their preference when comparing two differently-processed images side by side.

We collected data from 858 participants through a web-based study. After filtering based on response time patterns and incomplete responses, we retained 767 users. We define a user as a unique individual-device combination, as viewing conditions may influence preferences. Each user completed 90 pairwise comparison questions, yielding approximately 60,000 valid preference judgments after excluding cannot-choose responses. The dataset comprises 1,192 unique scenes processed into 7,972 unique image style pairs sourced from professional edits, device outputs, generative models, and synthetic transformations.

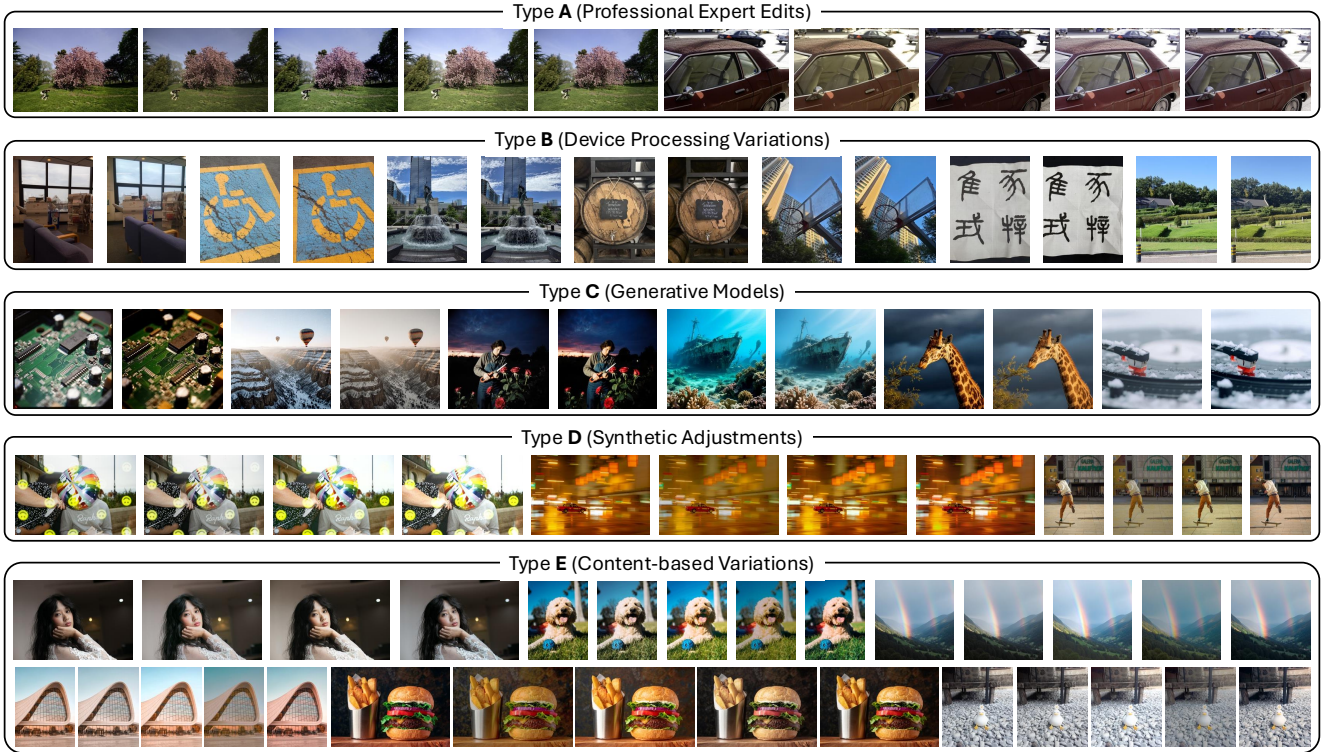


Figure 1. Sample images from each category in PPSD. **Type A** shows the same scene edited by different professional photographers. **Type B** presents the same scene captured by different camera devices. **Type C** displays generative model outputs with photo-realistic editing variations. **Type D** demonstrates controlled synthetic adjustments applied to the same base image. **Type E** exhibits content-based variations spanning diverse subjects. Each row within a type shows different style variations of the same or similar content.

### 3.2. Image Pair Construction

We designed PPSD to capture photographic style preferences across diverse sources of variation. The images are sourced from five distinct categories as shown in Figure 1.

**Professional Expert Edits (Type A).** We selected 188 scenes from the Adobe-MIT 5K dataset, where each scene includes multiple versions edited by different professional photographers with distinct aesthetic preferences, providing natural variations in color grading and tonal adjustments suitable for learning personalized style patterns.

**Device Processing Variations (Type B).** Different camera devices apply different image signal processing (ISP) pipelines to the same scene. We collected 308 image pairs across 12 camera devices to capture these variations. The collection includes 60 pairs from HKUST-ISP [57], 27 pairs from the ZRR [24] dataset, and 221 pairs that we captured across five device combinations including flagship smartphones from Samsung and Apple, as well as a Canon mirrorless camera. These pairs represent realistic style variations that users encounter when choosing between different devices or camera systems.

**Generative Model Outputs (Type C).** We created 161 pairs using FLUX-krea [7] and Qwen-Images [54] as base

generators. For each generated image, we applied a photo-realistic editing pipeline. First, we used the Llava [37] model to generate editing instructions that would produce photo-realistic style variations. We specifically instructed Llava to create scripts for photo-realistic edits rather than artistic or dramatic transformations. These instructions were then applied using FLUX-kontext [6], a widely-used image editing model, to produce the final edited versions. To our knowledge, this is the first use of FLUX-kontext for constructing a photo-realistic style transfer dataset. This category captures the diverse aesthetic space of modern generative models while maintaining photographic realism.

**Synthetic Adjustments (Type D).** We collected 121 base images from Pixabay and Unsplash, selecting photographically appealing images across diverse content without restricting to specific subject categories. We then applied controlled edits to create style variations: simple color manipulations (saturation reduction, brightness increase, and flash effects) and three professional LUTs. These controlled variations allow analysis of preferences for specific color and tone attributes in isolation.

**Content-based Variations (Type E).** To support content-aware approaches rather than uniform processing across



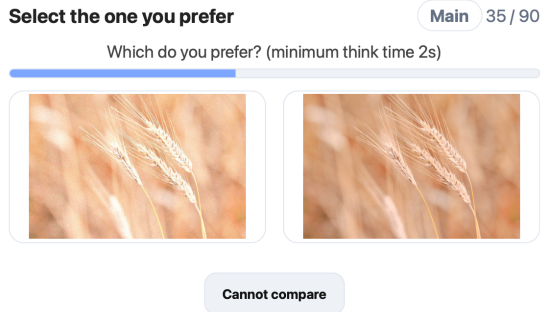


Figure 2. Web application interface for pairwise preference collection. Users select their preferred image or choose “Cannot compare” after a minimum 2-second viewing time.

all image types [32], we collected images spanning diverse photographic content. The collection includes 80 scenes from PPR10K, a professional retouching dataset. We generated five additional content-specific collections using FLUX-krea covering dogs, nature scenes, architecture, and food, totaling 327 scenes. We also included 46 scenes from the ORIDa dataset [29], which features the same physical object (a duck doll toy) captured under varying lighting and background conditions. Each collection includes multiple editing approaches from Type D applied to the images.

### 3.3. Data Collection using Web App

We developed a web application for collecting pairwise preferences. Participants first completed user registration and a pre-survey, received instructions on the task, and then proceeded to the main comparison questions. Details of each stage are provided in the supplementary material.

The main comparison task presented 90 questions per user: 80 primary comparisons and 10 consistency checks. In each question, two images appear side by side (Figure 2). For the 80 primary comparisons, we randomly assigned each user to 1-6 image categories from Section 3.2, and images were sampled randomly from the assigned categories. The 10 consistency checks reused image pairs from earlier in the session with left-right positions flipped, allowing for measuring the reliability of each user’s responses.

Users responded by selecting the left image, the right image, or indicating they cannot choose. To encourage thoughtful evaluation, each question required a minimum viewing time of 2 seconds. On average, users spent 5.8 seconds per comparison.

### 3.4. Dataset Statistics

We collected various statistics to characterize PPSD and its participants. Participants had an average age of 35.99 years and achieved a mean consistency score of 0.743 on the

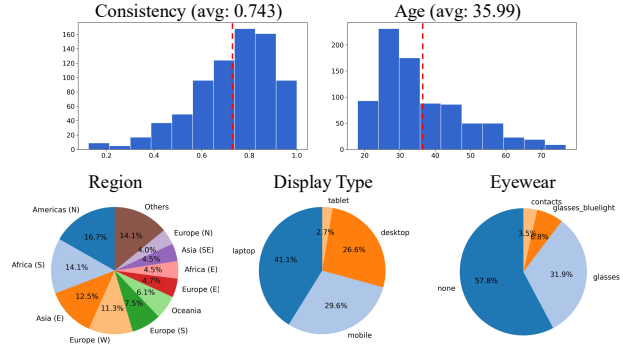


Figure 3. PPSD statistics showing distributions of user consistency scores, age, geographic regions, display device types, and eyewear usage. Best viewed zoomed in.

10 flipped pairs, indicating reliable preference expression. The average response time per comparison was 5.8 seconds. The dataset exhibits balanced gender representation (53.2% male, 46.1% female, 0.7% others) and predominantly normal color vision (744 normal, 12 with variations). Most participants reported good eyesight (85.1%), with left-side selection occurring at 51.4%, suggesting minimal positional bias. Additional statistics including geographic distribution, display device types, and participation across image categories are provided in Figure 3. The dataset represents diverse demographics collected globally. More statistics about PPSD are presented in the supplementary material.

## 4. Methodologies for PPS

In this section, our goal is to explore multiple models (Figure 4) as starting points for future research rather than developing a single model. Additionally, to address the challenges of learning from comparative preferences, we propose a training strategy using both preferred and non-preferred images, accompanied by an evaluation method that quantifies both target fidelity and preference alignment. **Training Framework.** We randomly select one image from a preference pair  $(I_p, I_n)$ , where the user preferred  $I_p$  over  $I_n$  for the same scene, as input and generate an output image. This prediction is supervised using both the preferred and non-preferred images, with details in Section 4.4. This training strategy applies to all three methods, though they differ in how they leverage the preference context.

**User-Specific Decoder.** This method pretrains a shared encoder to extract high-dimensional image features from large-scale preference data. At inference time, the encoder is frozen and a new decoder is initialized for each user. The decoder is trained on the user’s  $N$  reference preference samples using the same supervision strategy as pretraining. This inference-time training approach allows the decoder to specialize to individual photographic styles.

**User Preference Embedding.** This method employs a sim-

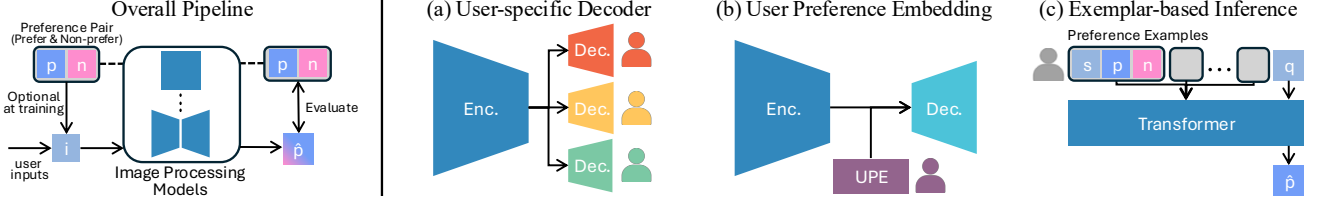


Figure 4. Overview of the inference pipeline and three baseline methods. **The overall pipeline** processes an input image (optionally obtained from user responses during training or user inputs at inference) through image processing models to generate personalized outputs with evaluation. **(a) User-Specific Decoder:** A shared encoder extracts features, while individual decoders capture personalized styles for each user. **(b) User Preference Embedding:** A global encoder-decoder conditioned on user preference embeddings (UPE) encoding individual aesthetic preferences. **(c) Exemplar-Based Inference:** A transformer processes preference triplets (scene, preferred, non-preferred images) to predict personalized style for a query image.

ilar encoder-decoder architecture but uses a single global decoder shared across all users. Personalization is achieved through user preference embeddings (UPE) that encode individual aesthetic preferences into compact vectors. During training, the decoder learns to condition on UPE to produce preference-aligned enhancements. At inference, a new user’s UPE is extracted from their  $N$  reference samples without additional training.

**Exemplar-Based Inference.** This method adapts in-context learning to our preference-based setting. We construct triplet-based examples consisting of (scene image, preferred image, non-preferred image), where the scene image is randomly selected from the pair with augmentation applied. During both training and inference,  $N$  reference triplets are provided as context to the transformer, which predicts the enhancement for a query image. Unlike settings with clear input-output pairs, our preference pairs lack explicit degradation. The triplet structure allows the model to learn the direction of preference from comparative examples, enabling preference-aligned stylization without inference-time training.

#### 4.1. User-specific Decoder

Our approach adopts an encoder-decoder architecture. The encoder  $E$  extracts features from the input image, while the decoder  $D$  transforms these features to produce the stylized output. For the decoder, we employ an implicit neural representation based on Local Implicit Image Function (LIIF) [10]. This formulation allows the decoder to predict pixel values at arbitrary spatial coordinates, providing flexibility in handling images of different resolutions and enabling fine-grained control over the transformation process.

Given an input image  $I \in \mathbb{R}^{H \times W \times 3}$ , the encoder extracts a feature map  $F = E(I) \in \mathbb{R}^{H \times W \times C}$ . For any normalized pixel coordinate  $c \in [0, 1]^2$ , the decoder predicts the RGB value  $\hat{I}(c) \in \mathbb{R}^3$  as:

$$\hat{I}(c) = D(F(c), c), \quad (1)$$

where  $F(c) \in \mathbb{R}^C$  denotes the feature vector at coordinate

$c$  obtained through spatial interpolation from  $F$ .

Our model maintains a single global encoder  $E$  shared across all users and individual decoders  $\{D_u\}_{u=1}^U$  for each of the  $U$  users (Figure 4(a)). For a given user  $u$  and input image  $I$ , the stylized output is generated as:

$$\hat{I}_u(c) = D_u(F(c), c), \quad \text{where } F = E(I). \quad (2)$$

#### 4.2. User Preference Embedding

This model employs an encoder-decoder architecture with implicit image functions as user-agnostic decoders (Figure 4(b)). For each user, we construct a UPE  $e_u \in \mathbb{R}^d$  from  $N$  preference examples. Specifically, given a set of  $N$  preference pairs  $\{(I_p^{(i)}, I_n^{(i)})\}_{i=1}^N$ , where  $I_p^{(i)}$  and  $I_n^{(i)}$  denote the preferred and non-preferred images for the same scene, we compute an embedding for each pair. For each pair  $i$ , the content embedding captures scene information by summing features, while the style embedding captures the preference direction by subtracting features:

$$e_{\text{content}}^{(i)} = f(I_p^{(i)}) + f(I_n^{(i)}), \quad (3)$$

$$e_{\text{style}}^{(i)} = f(I_p^{(i)}) - f(I_n^{(i)}), \quad (4)$$

where  $f(\cdot)$  is a feature extractor implemented using DINOv2 [44]. Each embedding is normalized before concatenation to form a pair-level embedding:  $e^{(i)} = [e_{\text{content}}^{(i)}; e_{\text{style}}^{(i)}]$ . The final UPE is obtained by aggregating the  $N$  pair-level embeddings through a fusion network:

$$e_u = g(\{e^{(i)}\}_{i=1}^N), \quad (5)$$

where  $g(\cdot)$  is implemented as a shallow Transformer encoder [52] that averages the output embeddings.

For a query image  $q$ , the encoder produces feature representation  $F_q$ , and the decoder generates the stylized output by conditioning on both the image features and the UPE:

$$\hat{I}_u(c) = D(F_q(c), e_u, c). \quad (6)$$

### 4.3. Exemplar-based Inference

We adapt PIE-MSM [32], a transformer-based approach that formulates image enhancement as a sequence prediction task. PIE-MSM employs a two-stage training strategy. In the first stage, an encoder-decoder network is trained for image-to-image translation. Given a source image  $I_s$  and target image  $I_t$ , the model extracts content and style representations through two encoders  $\phi_c(\cdot)$  and  $\psi_s(\cdot)$ :

$$z_{\text{content}} = \phi_c(I_s), \quad z_{\text{style}} = \psi_s(I_t) - \psi_s(I_s). \quad (7)$$

In the second stage, a transformer is trained to predict style embeddings from example pairs. Given enhancement examples  $\{(I_s^{(i)}, I_t^{(i)})\}_{i=1}^M$ , each pair is encoded as  $(z_{\text{content}}^{(i)}, z_{\text{style}}^{(i)})$  and fed to the transformer as context. At inference, given a query image  $q$ , its content embedding  $\phi_c(q)$  is appended to the context sequence, and the transformer predicts a style embedding  $\hat{z}_{\text{style}}^q$  for the query.

For PPS, we modify this framework to handle preference pairs without explicit input-output correspondence (Figure 4(c)). We provide  $N$  preference triplets as context:  $\{(s^{(i)}, p^{(i)}, n^{(i)})\}_{i=1}^N$ , where  $s^{(i)}$  is a scene image,  $p^{(i)}$  is the preferred image, and  $n^{(i)}$  is the non-preferred image. Each triplet is encoded as:

$$z_{\text{content}}^{(i)} = \phi_c(s^{(i)}), \quad (8)$$

$$z_{\text{prefer}}^{(i)} = \psi_s(p^{(i)}) - \psi_s(s^{(i)}), \quad (9)$$

$$z_{\text{non-prefer}}^{(i)} = \psi_s(n^{(i)}) - \psi_s(s^{(i)}). \quad (10)$$

Given a query image  $q$ , its content embedding  $\phi_c(q)$  is appended to the context triplets, and the transformer predicts a style embedding  $\hat{z}_{\text{prefer}}^q$  for generating the final stylization.

### 4.4. Training Objective

PPS learns from comparative preferences rather than explicit ground truth. A pairwise response indicates only that one image is preferred over another; neither image serves as an oracle or is incorrect. Photographic style is also highly sensitive, where subtle changes can yield unpleasing results.

We address these challenges by training with both preferred and non-preferred images. For a given preference pair  $(I_p, I_n)$  of the same scene, we compute losses  $\mathcal{L}_{\text{prefer}}$  and  $\mathcal{L}_{\text{non-prefer}}$  between the model’s prediction  $\hat{I}$  and both images. The total loss is formulated as:

$$\mathcal{L} = w \cdot \mathcal{L}_{\text{prefer}} + (1 - w) \cdot \mathcal{L}_{\text{non-prefer}}, \quad (11)$$

where  $w$  controls the balance between the two objectives, and  $\mathcal{L}_{\text{prefer}}$  and  $\mathcal{L}_{\text{non-prefer}}$  can be instantiated with standard loss functions such as  $\ell_1$ ,  $\ell_2$ , or perceptual losses. To handle the subtlety of photographic style, we employ a curriculum on  $w$ , beginning with  $w = 0.5$  and gradually increasing it during training. Starting with equal weighting helps the model learn the differences between the two styles before focusing more on the preferred one.

### 4.5. Comparative Quality Score

Evaluating PPS models requires capturing both prediction fidelity and alignment with user preferences. We propose an evaluation method that jointly measures these two aspects: how accurately predictions match their respective targets (fidelity) and how strongly predictions favor the preferred target over the non-preferred one (preference alignment).

Given a preference pair  $(I_p, I_n)$ , we apply a baseline method to both images as input, generating two outputs:  $\hat{I}_p$  (from input  $I_p$ ) and  $\hat{I}_n$  (from input  $I_n$ ). We then compute a metric  $\mathcal{M}$  between all combinations of predictions and ground truths:

$$\begin{aligned} d_{pp} &= \mathcal{M}(\hat{I}_p, I_p), & d_{pn} &= \mathcal{M}(\hat{I}_p, I_n), \\ d_{np} &= \mathcal{M}(\hat{I}_n, I_p), & d_{nn} &= \mathcal{M}(\hat{I}_n, I_n). \end{aligned} \quad (12)$$

Next, we aggregate these values to get the average model performance *against* the preferred target ( $\bar{d}_p$ ) and *against* the non-preferred target ( $\bar{d}_n$ ):

$$\bar{d}_p = \frac{d_{pp} + d_{np}}{2}, \quad \bar{d}_n = \frac{d_{pn} + d_{nn}}{2}. \quad (13)$$

To assess whether predictions are closer to the preferred target, a straightforward approach is to examine the difference  $\Delta = \bar{d}_p - \bar{d}_n$  (for  $\uparrow$  metrics). However, this measure alone is insufficient as it captures only relative preference alignment while ignoring absolute prediction quality. Two models with identical  $\Delta$  may have vastly different absolute  $\bar{d}_p$  and  $\bar{d}_n$  scores.

To address this limitation, we propose the Comparative Quality Score (CQS):

$$\text{CQS} = \text{BFS} \times (1 + \text{CMR}), \quad (14)$$

where the Base Fidelity Score (BFS) measures overall target correspondence and the Comparative Margin Ratio (CMR) quantifies the directional bias toward the preferred target.

The BFS is formulated as the geometric mean (or its reciprocal). This choice is critical, as it heavily penalizes models that have poor performance on *either* target, ensuring a good model must be reasonably faithful to both aesthetic styles. For metrics where lower is better (e.g.,  $\Delta E$ , LPIPS):

$$\text{BFS}_{\downarrow} = \frac{1}{\sqrt{\bar{d}_p \times \bar{d}_n}}, \quad \text{CMR}_{\downarrow} = \frac{\bar{d}_n - \bar{d}_p}{\bar{d}_n + \bar{d}_p}. \quad (15)$$

For metrics where higher is better (e.g., PSNR, SSIM):

$$\text{BFS}_{\uparrow} = \sqrt{\bar{d}_p \times \bar{d}_n}, \quad \text{CMR}_{\uparrow} = \frac{\bar{d}_p - \bar{d}_n}{\bar{d}_p + \bar{d}_n}. \quad (16)$$

The CMR term normalizes the “preference margin” ( $\bar{d}_p - \bar{d}_n$ ) by the total magnitude ( $\bar{d}_p + \bar{d}_n$ ), measuring relative preference alignment in a stable manner.

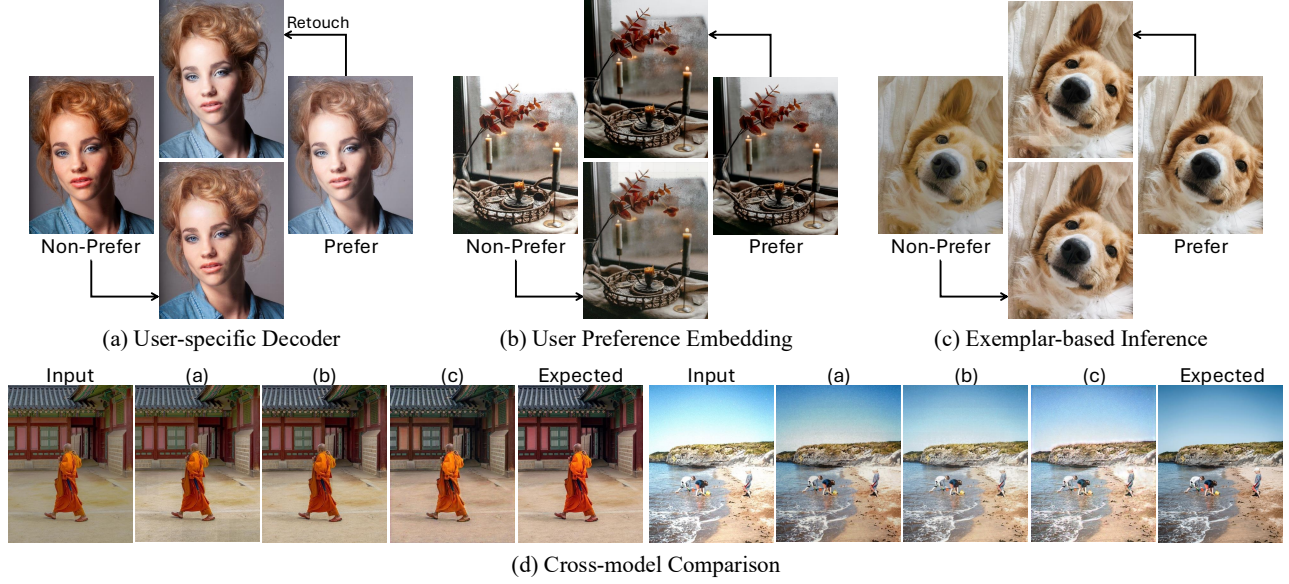


Figure 5. Qualitative results of baseline methods. **(a-c)** Given either non-preferred or preferred images as input, models adjust non-preferred images toward the user’s preferred aesthetic while preserving preferred ones. **(d)** Cross-model comparison on identical scenes, where outputs from different models processing the same input are shown between the input and expected result.

## 5. Experiment

### 5.1. Experimental Setup

**Dataset Split.** We filtered users with consistency scores below 0.7, retaining 521 users split into 471 for training and 50 for validation. For each validation user, we sample  $N = 16$  preference references and  $M = 16$  evaluation samples.

**Implementation.** For user-specific decoder and user preference embedding, we use EDSR [36] as encoder and HIIF [25] as decoder. For exemplar-based inference, we adopt the configuration from the official PIE-MSM repository. We train for 100 epochs. The implicit image function baselines use L1 loss. For PIE-MSM, we adapt the original loss formulation to our preference learning objective in Section 4.4. The weight curriculum parameter  $w$  in Equation 11 increases linearly from 0.5 to 1.5 over 500 epochs.

**Metrics.** We evaluate using multiple complementary metrics. For color accuracy, we use  $\Delta E_{00}$  [39] and  $\Delta E_{OKLAB}$  [45], which measure perceptual color differences. For perceptual similarity, we use LPIPS [60]. We also report PSNR and SSIM as pixel-level metrics.

### 5.2. Baseline Comparisons

**Qualitative Results.** Figure 5(a-c) shows models capturing distinct user preferences. Model (a) reduces saturation for a muted tone, model (b) adjusts brightness for a subdued atmosphere, and model (c) applies white balance correction and increases contrast. Figure 5(d) presents a cross-model comparison on identical inputs. While each model produces slightly different renditions, all three successfully move the

output closer to the user’s expected aesthetic. This alignment is evident in specific regions. For example, note the color rendition of the monk’s clothes, the building wall, and the ground in the left example, and the increased saturation of the sky and beach in the right example, all of which trend toward the expected result.

**Quantitative Results.** Table 1 reports results across multiple metrics. For evaluation, we use both preferred and non-preferred images as inputs, as models must learn when to adjust and when to preserve the original appearance.

The results clearly show Model (b) achieves the best performance, a finding captured by our proposed CQS. While Models (a) and (c) lag in fidelity, the positive CMR values across all methods are a key finding. This demonstrates that all baselines successfully learned to distinguish between preferences, validating the feasibility of the PPS task.

Moreover, we claim that the necessity of CQS over naive averages or deltas is evident in the  $\Delta E_{00}$  results. First, compare Model (a) and (c). Model (a) has a strong preference margin (CMR 0.050 vs 0.011), but its absolute fidelity is extremely poor (BFS 0.146 vs 0.196). Our CQS correctly penalizes this low fidelity, ranking (c) higher (0.197 vs 0.154). Second, compare Model (b) and (c). Here, Model (b) has slightly *worse* fidelity (BFS 0.189 vs 0.196). However, its preference alignment is far superior (CMR 0.078 vs 0.011). CQS correctly balances these factors, identifying that (b)’s strong preference alignment outweighs its minor fidelity gap, thus awarding it the top score (0.204). CQS is thus shown to be a robust metric that balances absolute fidelity (BFS) with preference alignment (CMR).



Table 1. Quantitative evaluation of three baseline approaches: Model (a) (User-specific Decoder), Model (b) (User Preference Embedding), and Model (c) (Exemplar-based Inference). Results are reported against preferred and non-preferred references (vs. Prefer, vs. Non-Prefer), along with BCS (Baseline Consistency Score), CMR (Comparative Match Rate), and CQS (Comparative Quality Score). Best and second-best CQS values per metric are **bolded** and underlined.

|                | $\Delta E_{00} \downarrow$ |              |              | LPIPS $\downarrow$ |             |           | PSNR $\uparrow$ |              |           | SSIM $\uparrow$ |              |              |
|----------------|----------------------------|--------------|--------------|--------------------|-------------|-----------|-----------------|--------------|-----------|-----------------|--------------|--------------|
|                | Model (a)                  | Model (b)    | Model (c)    | Model (a)          | Model (b)   | Model (c) | Model (a)       | Model (b)    | Model (c) | Model (a)       | Model (b)    | Model (c)    |
| vs. Prefer     | 6.50                       | 4.88         | 5.06         | 0.138              | 0.100       | 0.143     | 25.73           | 36.67        | 25.52     | 0.855           | 0.879        | 0.866        |
| vs. Non-Prefer | 7.18                       | 5.71         | 5.17         | 0.149              | 0.111       | 0.145     | 24.34           | 33.45        | 25.23     | 0.846           | 0.871        | 0.843        |
| BFS $\uparrow$ | 0.146                      | 0.189        | 0.196        | 6.97               | 9.50        | 6.95      | 25.03           | 35.02        | 25.37     | 0.851           | 0.875        | 0.867        |
| CMR $\uparrow$ | 0.050                      | 0.078        | 0.108        | 0.038              | 0.052       | 0.007     | 0.028           | 0.046        | 0.006     | 0.005           | 0.005        | 0.001        |
| CQS $\uparrow$ | 0.154                      | <b>0.204</b> | <u>0.197</u> | <u>7.24</u>        | <b>9.99</b> | 7.00      | <u>25.72</u>    | <b>36.63</b> | 25.52     | 0.855           | <b>0.879</b> | <u>0.867</u> |

Table 2. Ablation study on number of users for training.

| $\Delta E_{00}$ | N=50  | 200          | 471 (full)   | PSNR  | N=50  | 200          | 471 (full)   |
|-----------------|-------|--------------|--------------|-------|-------|--------------|--------------|
| vs P.           | 5.39  | 5.21         | 4.88         | vs P. | 30.4  | 32.2         | 36.7         |
| vs N.           | 5.94  | 5.99         | 5.71         | vs N. | 29.0  | 30.0         | 33.5         |
| CQS             | 0.185 | <u>0.192</u> | <b>0.204</b> | CQS   | 30.39 | <u>32.18</u> | <b>36.66</b> |

## 5.3. Ablation Studies

### 5.3.1. Effect of Training Set Size

We investigate how the number of training users affects personalization performance. Table 2 shows that performance consistently improves with more training users. This trend suggests that diverse user preferences help the model learn more generalizable personalization patterns. The continued improvement even at 471 users indicates that larger datasets could further enhance performance.

### 5.3.2. Preference Reference Composition

We analyze how the number and composition of preference references affect personalization quality (Table 3). Performance improves as  $N$  increases from 4 to 16, but declines at  $N = 32$ , suggesting that excessive references may introduce noise or conflicting signals. Among three selection strategies at  $N = 16$ , differences are subtle, with one performing slightly better. These results suggest that both reference quantity and selection strategy matter, while the optimal configuration may vary across users. Future work could explore adaptive selection methods or investigate whether reference scaling follows predictable patterns.

## 6. Discussion and Future Work

**Collecting Preferences Over Time.** Our data collection protocol limited each session to 90 pairwise comparisons, averaging 8.7 minutes per user, to account for perceptual fatigue and attentional constraints. While necessary for response quality, this concentrated format suggests opportunities for more natural collection paradigms. Mobile photography platforms could enable gradual preference accumulation through routine photo reviews, allowing users to provide feedback during everyday workflows rather than isolated sessions. Such approaches would support continuous refinement of personal style models without the constraints of concentrated collection events.

Table 3. Ablation study on number of reference images.

| $\Delta E_{00}$ | N=4   | 8     | 16a          | 16b   | 16c          | 32    |
|-----------------|-------|-------|--------------|-------|--------------|-------|
| vs P.           | 5.11  | 5.00  | 4.88         | 4.93  | 4.92         | 5.10  |
| vs N.           | 5.92  | 5.94  | 5.71         | 5.69  | 5.71         | 6.03  |
| CQS             | 0.195 | 0.199 | <b>0.204</b> | 0.202 | <u>0.203</u> | 0.195 |
| PSNR            | N=4   | 8     | 16a          | 16b   | 16c          | 32    |
| vs P.           | 34.7  | 35.7  | 36.7         | 36.6  | 36.7         | 33.0  |
| vs N.           | 32.2  | 33.0  | 33.5         | 33.8  | 34.1         | 30.4  |
| CQS             | 34.7  | 35.7  | <u>36.66</u> | 36.6  | <b>36.68</b> | 33.0  |

**Rethinking Fully Automated Image Retouching.** This work demonstrates technical feasibility of learning personalized photographic styles, however, an important question remains: “is fully automated enhancement the optimal user experience?” While automation reduces effort, people often value control in creative processes. The balance between automation and user agency requires further investigation, connecting technical capabilities with insights from human-computer interaction, user experience design, and cognitive psychology. Understanding how users prefer to interact with editing systems merits interdisciplinary exploration.

**Potential Methodological Directions.** Recent advances suggest natural methodological extensions. Diffusion models’ generative priors and preference optimization methods from natural language processing offer promising directions. Approaches such as Direct Preference Optimization [47] and its adaptations to image generation (DiffusionDPO [53], D3PO [58], and personalized variants [16]) have shown effectiveness in related domains. These methods remain largely confined to generation tasks, presenting opportunities for application to photographic style learning.

## 7. Conclusion

We introduced Personalized Photographic Style (PPS) learning, the task of capturing individual aesthetic preferences from comparative judgments. We presented PPSD, a large-scale pairwise preference dataset, and demonstrated the feasibility of PPS through empirical studies and a novel comparative evaluation framework. We hope this work provides a foundation for future research in personalized aesthetic photography.



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# Learning Personalized Photographic Style from Pairwise User Preferences

## Supplementary Material

### A. Web Application for Data Collection

We developed a web-based application to collect pairwise preference data at scale. This section describes the interface design and data collection procedures. The main survey procedure is described in Section 3.3 of the main paper.

#### A.1. Overall Procedure

The data collection process consists of five stages:

1. **Study Guide.** Participants review an overview explaining the study’s purpose, estimated completion time (10–20 minutes), and data privacy policy. The interface clarifies that responses are used for academic research and that aggregated results may be publicly released without personal identifiers.
2. **User Registration.** Participants provide demographic information and viewing environment details. An auto-generated user ID is assigned to each participant, which can be modified if needed. A seed number is also provided to enable reproducible comparison ordering for participants who wish to repeat the survey.
3. **Pre-Survey.** Before the main task, participants complete a display characterization survey to measure their viewing environment and color sensitivity. This stage captures display-specific properties and individual perceptual characteristics.
4. **Tutorial.** Participants review step-by-step instructions for the comparison task. The tutorial uses a sequential disclosure format where each instruction becomes visible only after acknowledging the previous one.
5. **Main Survey.** Participants complete 90 pairwise comparisons, selecting their preferred image based on color and tone. The survey enforces a minimum 2-second viewing time per pair to encourage careful evaluation. Details of this stage are provided in Section 3.3.

On average, participants completed the entire procedure in 11.6 minutes. This duration was designed to minimize visual fatigue and maintain attention throughout the survey.

#### A.2. User Registration

The registration form collects information about participants’ background. Participants specify their eyewear type (none, glasses, glasses with blue-light filter, or contact lenses), current vision quality (good, average, or poor), and color vision status (normal, red-green deficiency (protan), red-green deficiency (deutan), blue-yellow deficiency (tritan), achromatopsia, or unsure). They also indicate their display device type (desktop monitor, laptop, tablet, or mobile) and optionally provide the display model details.

**Study Guide**

In this short survey, you simply choose which of two images looks better in color/tone. There are no right answers — your choices help us understand general viewing preferences.

**Overview & Steps**

Choose based on color/tone differences between two images.

1. **Register:** Provide brief info to begin.
2. **Pretest & Tutorial:** A short pretest and a quick how-to tutorial.
3. **Main Session:** 90 comparisons.
4. **Finish:** Complete all comparisons and press Finish.

**Estimated Time**

About 10–20 minutes depending on your pace.

**Data & Privacy**

- ✓ Responses are used for academic/non-commercial research.
- ✓ Aggregated results or datasets may be released publicly (no personal identifiers).
- ✓ Your OS and browser info are automatically recorded to interpret results.
- ✓ No personally identifiable information is collected.

**I have read the above and agree to participate.**

Figure A. Capture image of study guide page (stage 1).

**User Registration**

Before starting, please provide the information below.

User ID \* U-19fb1400 Seed \* 1994336834

Auto-filled; you can change it. Use this to get the same order again. Unless you received separate instructions, you can ignore this.

Eyewear \* Current vision (with correction if used) \* Color vision \*

Display device \*

Choose the display you're using now.

Display model details (as specific as you can) e.g., iPhone 15 Pro / Galaxy S24 / LG 27UL650 / Dell U2720Q

Job / Field \* Region \* Gender \*

Age \*

**Start**

Figure B. Capture image of user registration page (stage 2).

Additionally, participants report their job or field, region, gender, and age group. The job field options include student, engineer, designer, researcher, artist, photographer, videographer, image editor, colorist, color scientist, display engineer, UI/UX designer, marketing professional, gamer, and others. In practice, most participants selected the “others” category, indicating diverse backgrounds.

### Pre-Survey

Please avoid heavy color adjustments (e.g., blue-light filters, vivid/boost modes). If possible, use a default/standard display mode.

Required: please review carefully before answering.

---

#### Near-black distinguishability

For each bar, select the single arrow at the first point where, along the dark→bright direction, the two sides of the arrow no longer differ in brightness.

---

#### Near-white distinguishability

For each bar, select the single arrow at the first point where, along the bright→dark direction, the two sides of the arrow no longer differ in brightness.

---

#### White point estimate

Use the slider below the color gradient to choose where your display white appears; pick the point that looks the most neutral/pure.

---

#### Gamma measurement

Adjust the right solid gray patch until it matches the average brightness of the left checkerboard. Viewing from farther away may help. Squint or slightly defocus your eyes to make matching easier. Gently moving your head side-to-side can also help.

2.20 v

---

#### Color tendency & basic gamut check (How does the gray look?)

Color of the gray patch

How does the gray look? #1 ☐ Neutral ☐ Reddish ☐ Bluish ☐ Greenish

How does the gray look? #2 ☐ Neutral ☐ Reddish ☐ Bluish ☐ Greenish

How does the gray look? #3 ☐ Neutral ☐ Reddish ☐ Bluish ☐ Greenish

Shape in red area

#1 ☐ Circle ☐ Triangle ☐ Not visible

#2 ☐ Circle ☐ Triangle ☐ Not visible

#3 ☐ Circle ☐ Triangle ☐ Not visible

Shape in blue area

#1 ☐ Circle ☐ Triangle ☐ Not visible

#2 ☐ Circle ☐ Triangle ☐ Not visible

#3 ☐ Circle ☐ Triangle ☐ Not visible

Figure C. Capture image of pre-survey page (stage 3).

### Tutorial before starting the experiment

After you click during the study, we immediately move to the next comparison. Trust your first impression.

Which do you prefer? (minimum think time 2.0s)

---

User study example

#### Pre-experiment checklist

Select the one you prefer

Photos are shown side by side. If you're on a vertical display (e.g., phone) and the photos feel too small, please rotate your device to landscape.

Focus primarily on color/tone differences.

Please avoid heavy color adjustments (e.g., blue-light filters, vivid/boost modes). If possible, use a default/standard display mode.

A minimum think time of 2s is enforced before clicking.

Clicking an image instantly loads the next pair.

Use 'Cannot compare' when unsure, or when the two photos are clearly different images (different content or very different composition).

Some scenes may appear more than once during the survey.

If you stop mid-way, your data will be excluded from aggregation.

Participation counts after completing all comparisons and pressing Finish.

We also record total time spent during the experiment for analysis.

Figure D. Capture image of tutorial page (stage 4).

### A.3. Pre-Survey

The pre-survey characterizes each participant's display and color perception. This stage consists of four measurements:

- **Near-Black Distinguishability.** Participants view a series of gradually darkening bars and identify the point where brightness differences become indistinguishable. This measures the display's black level and the participant's sensitivity to dark tones.
- **Near-White Distinguishability.** Using a similar approach with brightening bars, this test measures the display's white level handling and the participant's sensitivity to highlight details.
- **White Point Estimate.** Participants select a point on a color gradient that appears most neutral or pure white on their display. This captures the display's white point characteristics and any color tinting present in the viewing environment.
- **Gamma Measurement.** Participants adjust a solid gray patch to match the average brightness of a checkerboard pattern. This estimates the effective gamma curve of their display system, which affects how brightness levels are distributed across the tonal range.



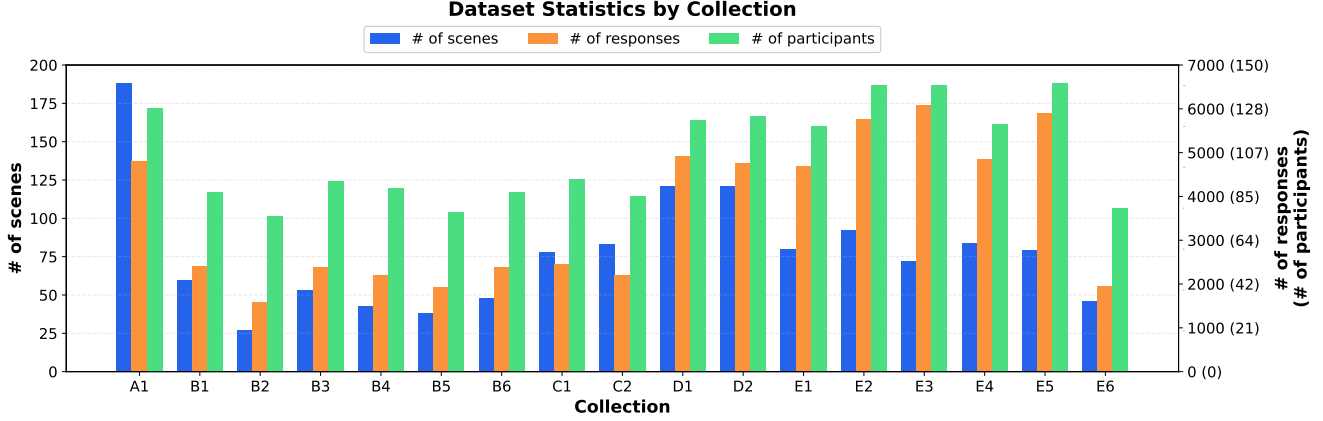


Figure E. **Dataset statistics by collection.** Distribution of scenes (blue bars), responses (orange bars), and participants (green bars) across all 17 subcollections in PPSD. The right y-axis shows the number of responses and participants. Collection A1 contains the most scenes (188), while content-specific collections (E2-E5) attracted the highest participation with over 6,000 responses each. The dataset achieves broad coverage across all collection types, with each receiving substantial user engagement.

- **Color Tendency and Basic Gamut Check.** Participants evaluate three gray patches and indicate whether each appears neutral or exhibits a color cast (reddish, bluish, or greenish). They also identify shapes within solid red and blue areas. These tests assess color tinting in the display and the participant’s sensitivity to subtle color variations.

While these measurements were not directly used in the baseline experiments presented, they are included in the released dataset. Analysis of how personal display settings and color sensitivity correlate with aesthetic preferences represents a promising direction for future work.

#### A.4. Tutorial

The tutorial stage ensures participants understand the comparison task and viewing guidelines. Instructions are presented sequentially, with each item revealed only after the participant acknowledges understanding of the previous one. This design prevents participants from skipping important information and ensures all guidelines are reviewed before beginning the main survey.

### B. Dataset Statistics

#### B.1. Details on Image Collections

Our dataset comprises five main categories (Types A through E) as described in Section 3.2. Here we provide additional details on the subcollections within each type and analyze their distribution in the dataset.

**Collection Structure.** Each main type is further divided into subcollections based on specific sources or processing methods. Type A consists of a single collection (A1) from Adobe-MIT 5K [8]. Type B is divided into six subcollections (B1-B6) representing different device pairs or dataset sources: B1 from HKUST-ISP dataset [57], B2 from the

Raw-to-Raw [24] dataset, and B3-B6 from our own captures using various smartphone and camera combinations (Galaxy S25 Ultra vs. iPhone 15 Pro, Galaxy S24+ vs. Galaxy S20, Galaxy Note10 vs. iPhone 16 Pro, Canon EOS R50 vs. iPhone 16+). Type C includes two subcollections from different generative models: C1 using FLUX-krea [7] and C2 using Qwen-Images [54]. Type D is split into two subcollections: D1 with simple color manipulations (four style variations per scene) and D2 with LUT-based retouching (four LUTs applied per scene). Type E is divided into six content-specific subcollections (E1-E6), where E1 uses PPR10K images and E2-E6 focus on specific subjects (dogs, nature, architecture, food, and duck dolls, respectively). For E2 through E6, each base image generated by FLUX-krea was processed with four consistent retouching approaches: saturation reduction, brightness increase, and two different LUTs.

**Dataset Statistics by Collection.** Figure E presents the distribution of scenes, responses, and participants across all subcollections. During data collection, each participant was randomly assigned 1 to 6 collections from the 17 available subcollections (A1-E6). Collection A1 contains the largest number of scenes (188) and received over 6,000 responses from 128 participants. Among device-based collections (B1-B6), the distribution varies based on source availability, with B1 from HKUST-ISP containing 60 scenes and our self-captured collections (B2-B6) ranging from 27 to 53 scenes. The generative model collections (C1-C2) show balanced coverage with 78 and 83 scenes respectively. The controlled synthetic collections (D1-D2) each contain 121 scenes. Content-specific collections (E2-E6) received substantial engagement, with E3 (nature) gathering over 6,000 responses from 138 participants. The variation in response counts across collections reflects both the number of

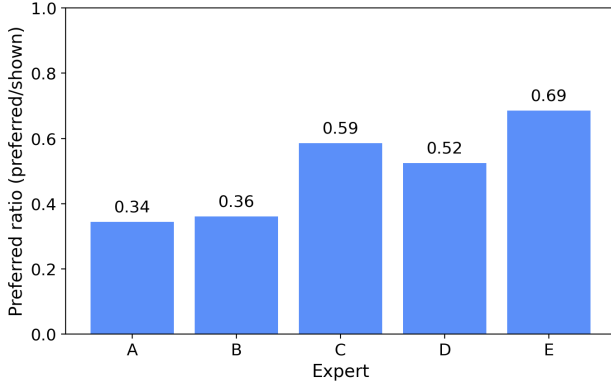


Figure F. Preferred ratio for each Adobe-MIT 5K expert style.

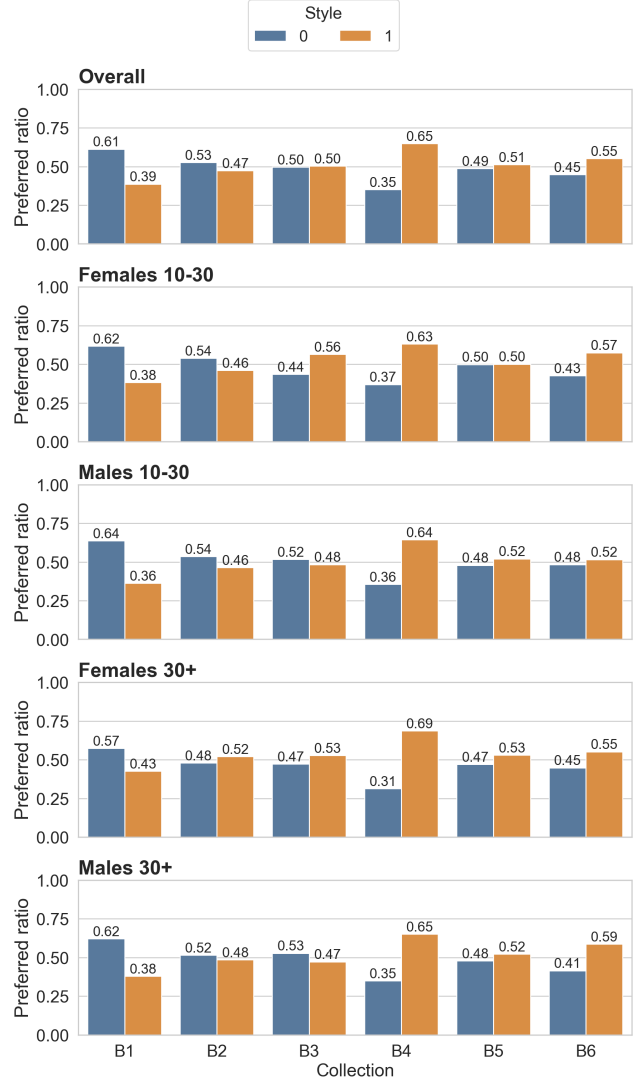
scenes and the random assignment of collections to participants. Overall, the dataset achieved broad coverage across all collection types, ensuring diverse representation of photographic style variations.

## B.2. Preference Pattern Analysis

We analyze several interesting patterns that emerge from user responses. This analysis provides insights into how different retouching sources and styles are perceived across our user base.

**Which Adobe-MIT 5K expert is most preferred?** The Adobe-MIT 5K dataset provides five expert retouching results (A, B, C, D, E) for each image. We examine which expert’s aesthetic receives the highest preference rate when compared against other experts or alternative retouching methods. Figure F shows the preferred ratio for each expert style in our dataset, where the ratio is calculated as the number of times each style was preferred divided by the number of times it was shown to users. Expert E (style 4) shows the highest preference rate at 0.69, followed by Expert C (style 2) at 0.59, while Experts A and B (styles 0 and 1) show lower rates of 0.34 and 0.36. This could suggest that certain expert aesthetics tend to resonate more with our user base. However, we note that these observations are based on a subset of Adobe-MIT 5K images, and preference rates may vary depending on which style pairs are presented together.

**Galaxy vs. iPhone: Do young women really prefer iPhone captures?** Different smartphone manufacturers employ distinct image processing pipelines, which may lead to varying user preferences. We examine whether demographic factors such as age and gender influence preferences for images captured with Samsung Galaxy and Apple iPhone devices. Figure G shows preferred ratios across six device pair collections (B1-B6), where style 0 and style 1 represent different device outputs in each collection. The overall results show varied preferences depending on the



|   | B1         | B2        | B3               | B4          | B5             | B6            |
|---|------------|-----------|------------------|-------------|----------------|---------------|
| 0 | iPhone (?) | Galaxy S9 | Galaxy S25 Ultra | Galaxy S24+ | Galaxy Note 10 | Canon EOS R50 |
| 1 | Mi (?)     | iPhone X  | iPhone 15 Pro    | Galaxy S20  | iPhone 16 Pro  | iPhone 16+    |

Figure G. Preferred ratios for individual devices by age and gender.

specific device pair, with no universal dominance of either manufacturer. However, demographic analysis reveals interesting patterns. In collection B2 (Galaxy S9 vs. iPhone X), females show a notably higher preference for iPhone captures (0.54) compared to males (0.54 vs. 0.46), and this preference is strongest among younger females (10-30: 0.54 vs. 30+: 0.48). Within the female group, younger users consistently show higher iPhone preference rates across multiple collections. These patterns suggest that demographic factors may influence device preference, though the extent varies by specific device models.

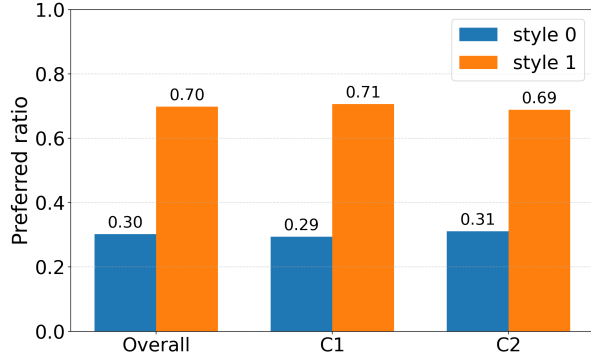


Figure H. Preferred ratio for generated image before and after generative editing.

**Generated images: Initial output vs. refined edits** For images created by generative models, we compare user preferences between the initial generated output and subsequently edited versions. This reveals whether users favor the original generative aesthetic or prefer additional refinements. Figure H shows preferred ratios for generated images before (style 0) and after (style 1) editing. Across both collections (C1: Flux-krea, C2: Qwen-Image), the edited versions consistently receive higher preference rates, with overall rates of 0.70 and 0.69 for edited images compared to 0.30 and 0.31 for initial outputs. In our pipeline, edited versions are created by extracting prompts from initial outputs using LLaVA and regenerating with Flux-kontext [6]. This suggests that users tend to prefer refined outputs over initial generations, regardless of the base generation model. Future work could explore editing capabilities of recent model-consistency-preserving approaches such as Nano Banana [14, 20] and other specialized editing models [3, 42, 49].

**Are there universally preferred synthetic retouching styles?** We analyze whether certain synthetic retouching approaches (e.g., specific filter combinations, adjustment patterns) consistently receive higher preference rates across diverse users and image types, or if preferences remain highly individualized. Figure I shows preferred ratios for two collections (D1, D2) with identical scene content, where style 0 represents the original image from public repositories (Pixabay [1], Unsplash [2]) or generation models. In collection D1, we apply synthetic edits: reduced saturation (style 1), increased brightness (style 2), and a flash effect with center brightness enhancement and noise (style 3). Collection D2 applies different public LUTs (4K-ALEXA-LOG-C-LTL.cube, ot\_1.A032.07211259.C029.cube,

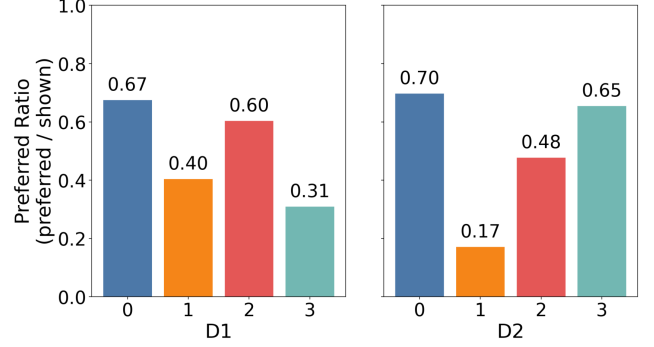


Figure I. Preferred ratios for original images and synthetic retouching styles.

SATICI.PIRABOK.1.C0026.cube) to styles 1, 2, and 3 respectively. Both collections show high preference for original images (0.67 in D1, 0.70 in D2), suggesting that images from public sharing platforms maintain good quality. In D1, brightness enhancement (style 2) achieves a comparable preference rate (0.60) to the original, while saturation reduction (style 1: 0.40) and flash effects (style 3: 0.31) receive lower preferences. This indicates that certain global adjustments may align better with general preferences. For D2, the original image receives the highest preference, suggesting that scene-agnostic LUT applications provide limited aesthetic improvement. However, style 3 shows a notably higher preference rate (0.65) compared to styles 1 (0.17) and 2 (0.48), indicating that some LUTs may be more universally appealing than others.

**Do semantic categories influence retouching preferences?** We examine whether image content (e.g., animals, landscapes, architecture, food) correlates with specific retouching preference patterns. Figure J shows preferred ratios across four semantic categories: animals (E2), landscapes (E3), architecture (E4), and food (E5). All images are generated using Flux-krea (style 0) and then edited with saturation reduction (style 1), brightness enhancement (style 2), LUT1 (4K-ALEXA-LOG-C-LTL.cube, style 3), and LUT2 (ot\_1.A032.07211259.C029.cube, style 4). Across all categories, we observe general tendencies: original images and brightness-enhanced versions receive higher preferences, LUT2 is generally preferred over LUT1, and saturation reduction consistently receives lower preference rates. Interestingly, for food images (E5), the preference against saturation reduction intensifies, while brightness enhancement receives substantially higher preference. This may reflect the importance of color saturation in conveying food freshness and appeal. These patterns suggest that while certain retouching tendencies exist across categories, semantic content can influence the degree of preference for specific adjustments.

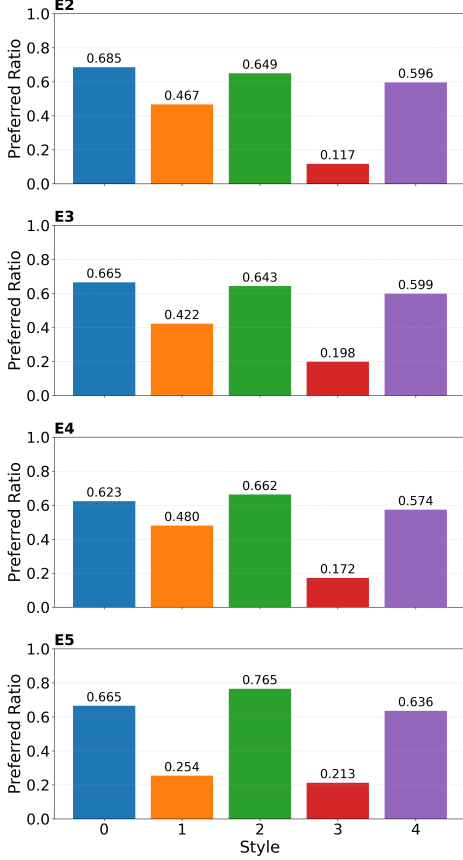


Figure J. Preferred ratios for styles by semantic category.

## C. Additional Experiments

This section presents additional empirical results for the proposed baseline methods. We note that our work focuses on establishing a foundation for personalized photographic style learning, demonstrating proof-of-concept for different technical approaches that can be further developed.

### C.1. Ablation Study on Training Weight

Table A shows the impact of training weights  $w$  assigned to preferred and non-preferred samples. Comparing (0.5, 1.0) and (0.5, 1.5), both configurations produce outputs closer to preferred images, as indicated by the vs P. metrics. The (0.5, 1.0) setting achieves better BFS scores (0.189 vs 0.191 for  $\Delta_{01}$ ), suggesting stronger preference alignment. However, (0.5, 1.5) shows better CMR performance (0.078 vs 0.054) and achieves higher CQS (0.204 vs 0.202). Since CQS balances both aspects of quality, we adopt (0.5, 1.5) as the default. When increasing the weight on preferred samples to (1.0, 2.0), CMR improves further (0.050 for  $\Delta_{00}$ ), but BFS degrades notably (0.167). This results in lower CQS (0.182), indicating that over-emphasizing pre-

Table A. Impact of training weights on model performance.

| $\Delta E_{00}$ | w=(0.5, 1.5) | (0.5, 1.0) | (1.0, 2.0) |
|-----------------|--------------|------------|------------|
| vs P.           | 4.88         | 4.95       | 5.47       |
| vs N.           | 5.71         | 5.52       | 6.55       |
| BFS             | 0.189        | 0.191      | 0.167      |
| CMR             | 0.078        | 0.054      | 0.090      |
| CQS             | <b>0.204</b> | 0.202      | 0.182      |

| PSNR  | w=(0.5, 1.5) | (0.5, 1.0) | (1.0, 2.0) |
|-------|--------------|------------|------------|
| vs P. | 36.7         | 36.5       | 32.7       |
| vs N. | 33.5         | 34.5       | 29.6       |
| BFS   | 35.05        | 35.49      | 31.11      |
| CMR   | 0.046        | 0.028      | 0.050      |
| CQS   | <b>36.66</b> | 36.49      | 32.66      |

Table B. Quantitative results on individual image subsets.

| $\Delta_{00}$ | Type A | B     | C     | D     | E     |
|---------------|--------|-------|-------|-------|-------|
| vs P.         | 4.54   | 6.78  | 5.26  | 3.92  | 3.77  |
| vs N.         | 4.95   | 7.25  | 5.64  | 4.72  | 4.35  |
| BFS↑          | 0.211  | 0.143 | 0.184 | 0.242 | 0.247 |
| CMR↑          | 0.043  | 0.034 | 0.035 | 0.052 | 0.071 |
| CQS↑          | 0.220  | 0.147 | 0.190 | 0.254 | 0.264 |

| PSNR  | Type A | B     | C     | D     | E     |
|-------|--------|-------|-------|-------|-------|
| vs P. | 36.65  | 34.63 | 37.15 | 37.38 | 37.72 |
| vs N. | 34.91  | 33.07 | 36.19 | 33.91 | 34.25 |
| BFS↑  | 35.77  | 33.84 | 36.67 | 35.60 | 35.94 |
| CMR↑  | 0.024  | 0.023 | 0.013 | 0.049 | 0.048 |
| CQS↑  | 36.64  | 34.62 | 37.15 | 37.33 | 37.67 |

ferred samples can harm overall performance. These results suggest that balanced weighting between preferred and non-preferred samples is important for learning personalized photographic style.

### C.2. Results on Individual Subsets

We evaluated our model across the five subsets (Types A-E). Table B shows distinct performance characteristics that offer important insights. We observed that the model finds Type B (Cross-device) the most challenging. This suggests that learning preferences from pairs with significant sensor and ISP discrepancies remains a difficult task. On the other hand, the model achieved high performance on Types D and E, indicating that preferences derived from synthetic color transformations are comparatively easier to learn.

### C.3. Human Evaluation

We conducted a study with 21 participants (Figure K). Across all models, participants consistently preferred the retouched results over the non-preferred originals, providing direct evidence that training on PPSD enables models to produce user-preferred photographic styles. Notably, the retouched images were also often comparable to the preferred originals, even when retouching starts from a non-preferred image. Finally, the human preference trend aligns with our



|           | Non-prefer vs. Retouch from non-prefer |     | Non-prefer vs. Retouch from prefer       |     | Prefer vs. Retouch from non-prefer |     | Prefer vs. Retouch from prefer       |     |
|-----------|----------------------------------------|-----|------------------------------------------|-----|------------------------------------|-----|--------------------------------------|-----|
| Model (a) | 26%                                    | 74% | 30%                                      | 70% | 67%                                | 33% | 68%                                  | 32% |
| (b)       | 20%                                    | 80% | 22%                                      | 78% | 48%                                | 52% | 44%                                  | 56% |
| (c)       | 29%                                    | 71% | 28%                                      | 72% | 58%                                | 42% | 45%                                  | 55% |
|           | Non-preferred image                    |     | Retouched image from non-preferred image |     | Preferred image                    |     | Retouched image from preferred image |     |

Figure K. Human evaluation study results. We conducted a pairwise preference study with 21 participants to evaluate Models (a-c).



Figure L. Content-aware preference application for user gnBuxFL-desktop. The model (User Preference Embedding model) applies dimming to architecture and nature (left) while brightening food images (right).

main-paper CQS-based evaluation: Model (b) is most preferred, and Model (c) is slightly stronger than Model (a) while remaining broadly comparable, supporting CQS as a meaningful proxy for pair-wise comparison quality.

#### C.4. Per-User Retouching Variations

Figure L demonstrates how the trained models apply learned user preferences in a content-aware manner. The left column shows cases where the model applies dimming effects to images of architecture and nature scenes. The bridge image becomes darker with enhanced contrast, while the cherry blossom photo shifts toward muted tones. The right column shows the same user’s model applying opposite adjustments to food subjects. The ramen bowl receives warmer tones and increased brightness, and the bread images are enhanced with stronger highlights.

These examples illustrate that the model does not learn a single global adjustment. Instead, it adapts the user’s aesthetic preferences based on the content and context of each image. This content-aware behavior suggests that our baseline approaches can capture meaningful aspects of personal photographic style when trained on sufficient comparative preference data.

#### C.5. Additional Qualitative Results

Figure M(a-c) shows additional results from the three baseline approaches across diverse image types. When starting from non-preferred inputs, all methods produce outputs that move toward the preferred aesthetic. When starting from preferred inputs, the models tend to maintain the preferred color and tone rather than deviating significantly.

It is worth noting that outputs from non-preferred inputs may still remain closer to the non-preferred appearance than to the preferred ground truth. However, the important observation is that the models consistently shift in the direction of the preferred style. This demonstrates that the baseline approaches can learn meaningful aspects of personal preferences from comparative data.

Figure M(d) compares all three baseline methods on the same inputs. Different methods show varying degrees of success across images. For some cases, one method produces results closer to the expected output, while for others, a different method achieves better alignment. This variability reflects the difficulty of this problem. These results establish baseline capabilities while highlighting opportunities for methodological improvements.

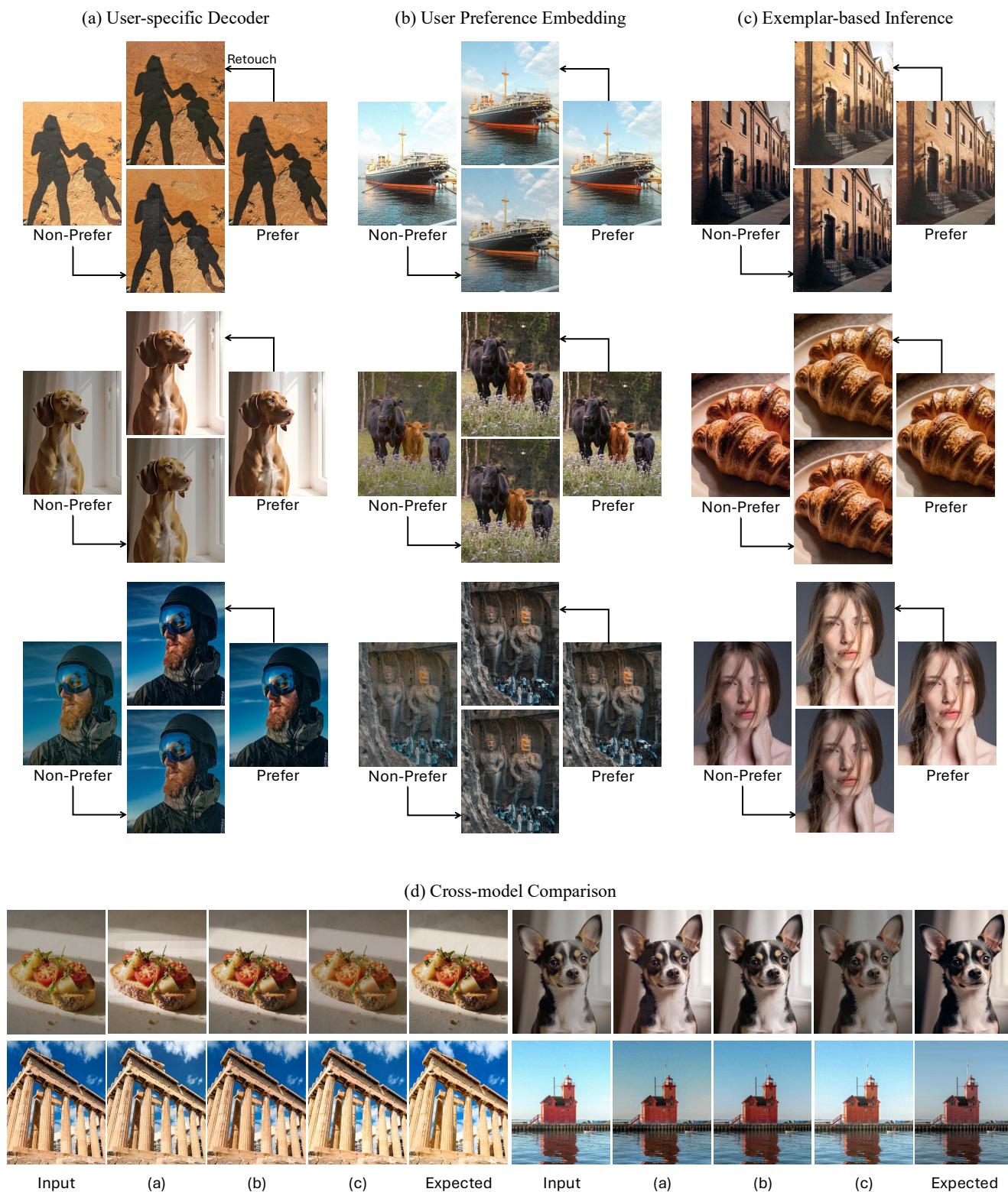


Figure M. Additional qualitative results for baseline methods. (a-c) Outputs from non-preferred and preferred inputs. (d) Cross-model comparison showing performance variation across different approaches.



## D. Limitation and Discussion

**Dataset Scale and Temporal Coverage.** Our dataset captures an average of 70 pairwise comparisons per user, collected within a single session. While this scale demonstrates the feasibility of collecting personalized preference data and enables meaningful baseline training, it represents a limited snapshot of each user’s aesthetic preferences. Personal style may require more extensive observation to fully characterize, particularly when preferences evolve over time or vary across different photographic contexts. Ideally, preference data would be collected continuously over extended periods, capturing how individual aesthetics develop and stabilize. Our work establishes a foundation at an academic scale, opening this research direction for future efforts.

**Display-Agnostic Preference Modeling.** The perception of photographic style is inherently tied to display characteristics. The same image array can produce different aesthetic experiences depending on display technology, calibration, and viewing conditions. While we made efforts to account for display factors by defining unique users as (user, display\_type) pairs and collecting display information in our pre-survey, our approach does not fully address display-centric aspects of PPS. Furthermore, we do not capture dynamic display adaptations such as True Tone on iOS devices or adaptive color tone features on Galaxy devices, which adjust rendering based on ambient conditions. Understanding how display settings interact with learned preference models remains an important consideration for future work.

**Proof-of-Concept Modeling Approaches.** Our baseline methods demonstrate the viability of different learning paradigms for PPS but are not intended as optimized solutions. Each approach reveals different trade-offs and potential directions, yet we did not pursue a single, comprehensive architecture that fully addresses the challenges of this problem. Developing more sophisticated models that effectively integrate preference signals across varied photographic contexts remains an open challenge.

**Evaluation Without Direct User Response.** Our work establishes a reusable evaluation framework based on real-world user responses. While this method effectively validates performance against recorded preferences, we recognize the value of confirming results through live user interaction. Future work can build on this foundation by developing sustainable evaluation methods that are both low-cost and reproducible. We envision a system where offline validation is seamlessly integrated with streamlined human feedback loops.

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